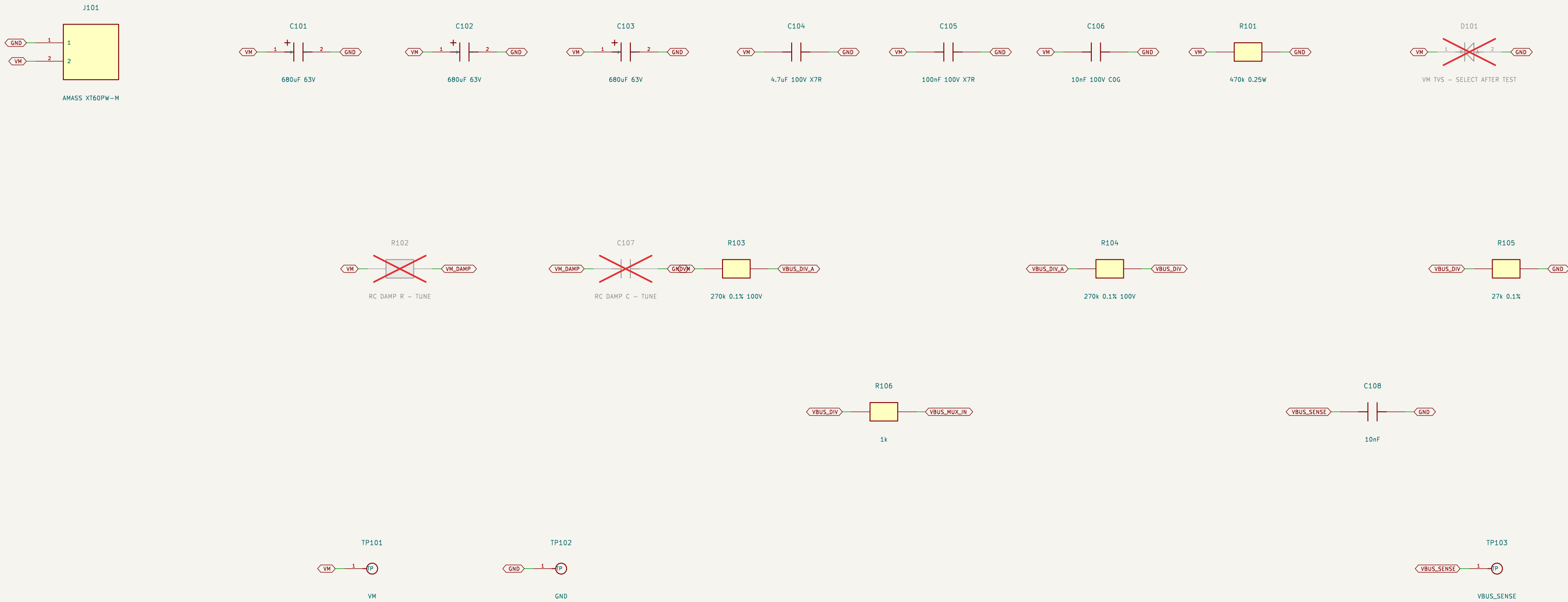


	1	2	3	4	5	6	7	8
A	ShiroFOC A0 – PCB layout release							
	18–42 V bus target 3–phase inverter STSPIN32G4 3 x Kelvin current sense							
	USB–priority control power CAN onboard encoder + external ABI/Hall BMI323							
	Read ../DESIGN_REVIEW_REV_A.md and ../PCB_LAYOUT_HANDOFF.md before placement.							
B	01 – Battery Input and DC Link				02 – Auxiliary Power, USB and Source Mux			
	File: 01_Battery_DC_Link.kicad_sch				File: 02_Aux_Power_USB.kicad_sch			
C	03 – STSPIN32G4 Core and Gate–Driver Supply				04 – Three–Phase Inverter and Current Sensing			
	File: 03_STSPIN_Core.kicad_sch				File: 04_Inverter_CurrentSense.kicad_sch			
	05 – Brake Chopper				06 – Onboard and External Position Feedback			
	File: 05_Brake_Chopper.kicad_sch				File: 06_Position_Feedback.kicad_sch			
D	07 – IMU and Half–Bridge Temperature				08 – CAN / CAN–FD Interface			
	File: 07_IMU_Temperature.kicad_sch				File: 08_CAN.kicad_sch			
E	09 – USB–UART Service Interface				10 – SWD, Boot and Test Access			
	File: 09_USB_UART.kicad_sch				File: 10_SWD_Test.kicad_sch			
F	1	2	3	4	5	6	7	8

BATTERY INPUT / DC LINK

Rev A: keyed XT60, no onboard reverse-polarity or precharge stage.



PLACEMENT: keep all three 680 uF capacitors close to the bridge bus. Fit local 1 uF + 10 nF / 100 V ceramics at each power block.

R102/C107 and D101 are DNP tuning positions. Select only after cable/hot-plug measurements; they are not functional requirements for first power-up.

VBUS_SENSE = VM x 27k/(540k+27k): 2.00 V at 42 V, 2.86 V at 60 V. Set the firmware overvoltage trip below the hardware transient ceiling.

ERC sources: connector input / passive-filtered regulator outputs / Internal bootstrap supplies.

GND is common; preserve Kelvin and power-return routing
PCB layout release; prototype ratings require hardware validation
ShiroFOC

Sheet: /01 - Battery Input and DC Link/
File: 01_Battery_DC_Link.kicad_sch

Title: 01 - Battery Input and DC Link

Size: A2

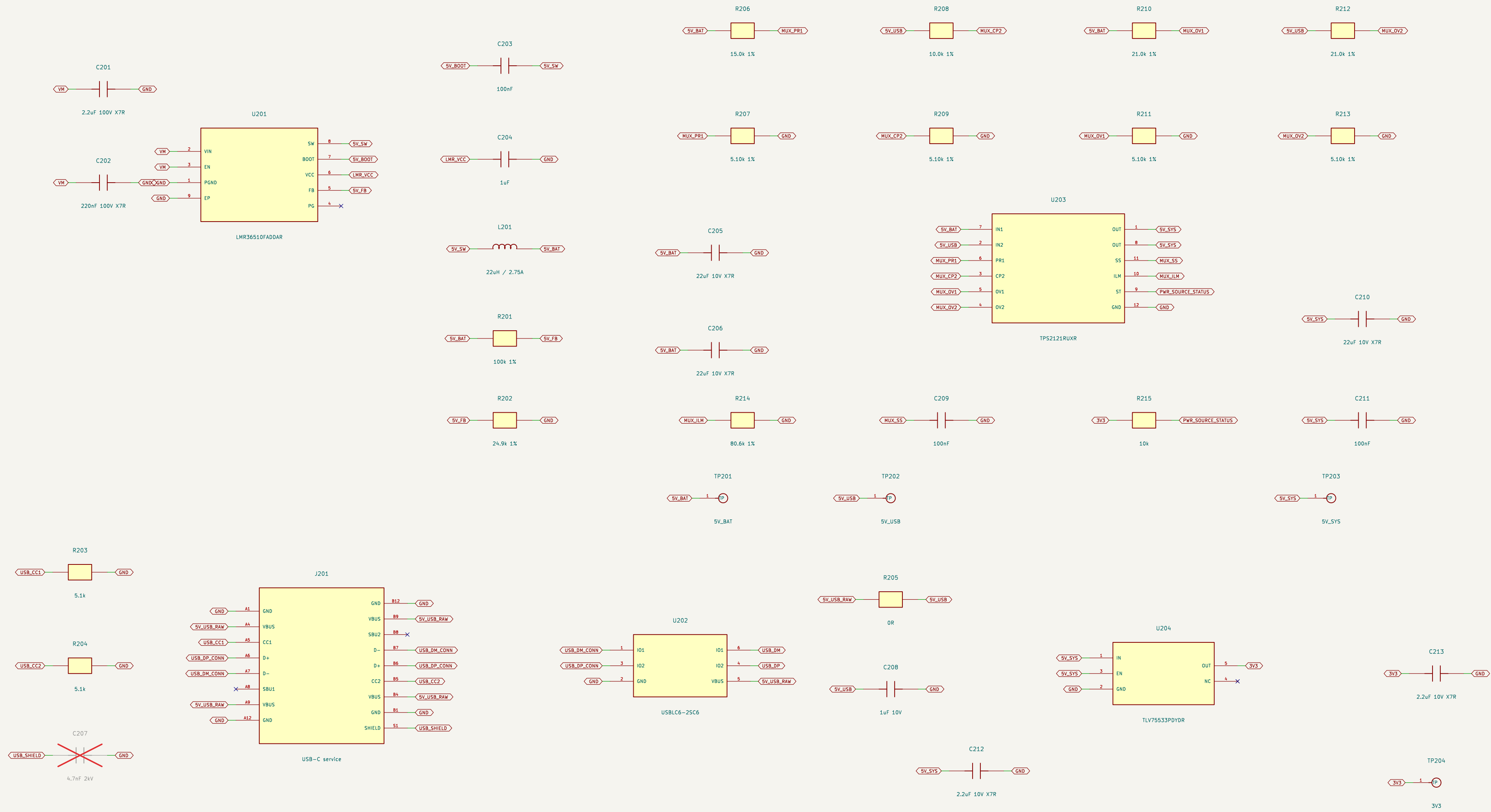
Date: 2026-09-15

Rev: A0

KiCad E.D.A. 9.0.4

Id: 2/11

VM -> 5V_BAT; USB VBUS -> 5V_USB; USB has priority; 5V_SYS -> 3V3



USB priority: PR1=1.269 V and CP2=1.689 V at 5 V. ILM=80.6k sets approx. 1.5 A. OV dividers trip near 5.43 V typical.

The TLV755 has no reverse-current blocking; SWD VTREF is sense-only; never inject 3V3 from the debugger.

ERC sources: connector input / passive-filtered regulator outputs / internal bootstrap supplies.

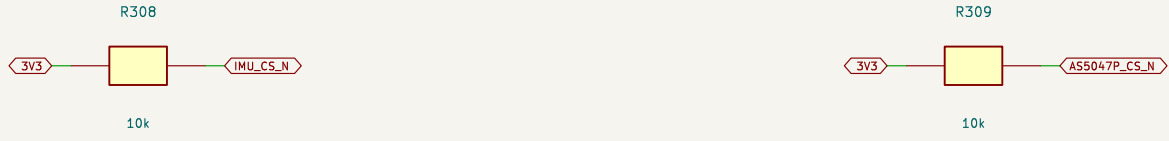
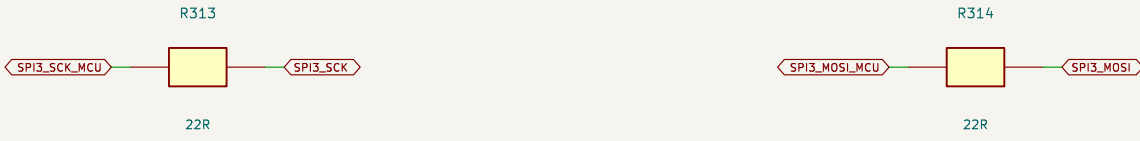
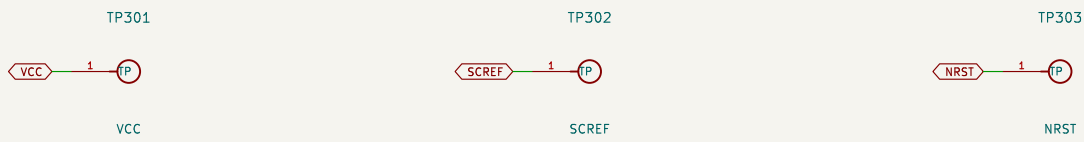
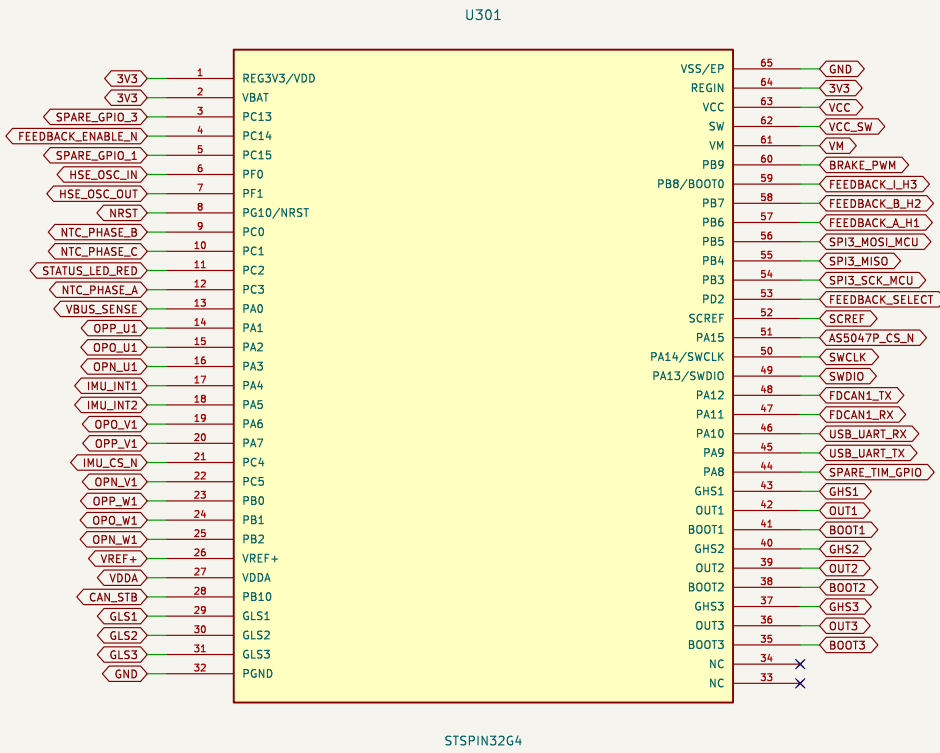
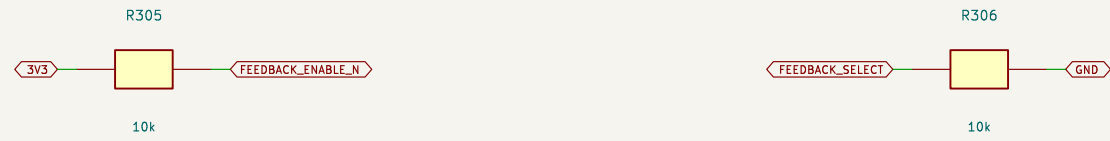
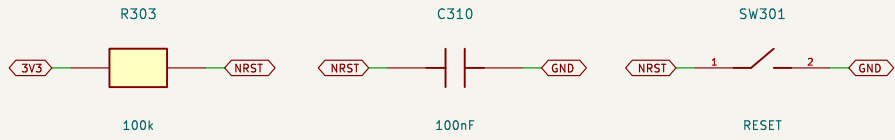
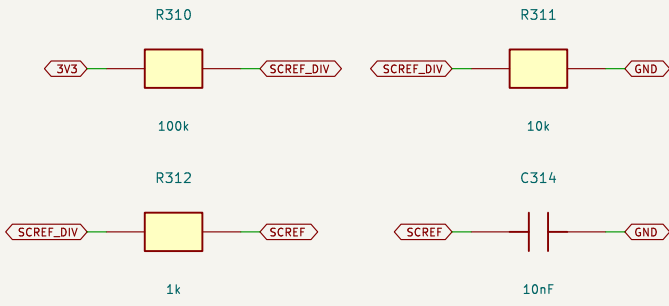
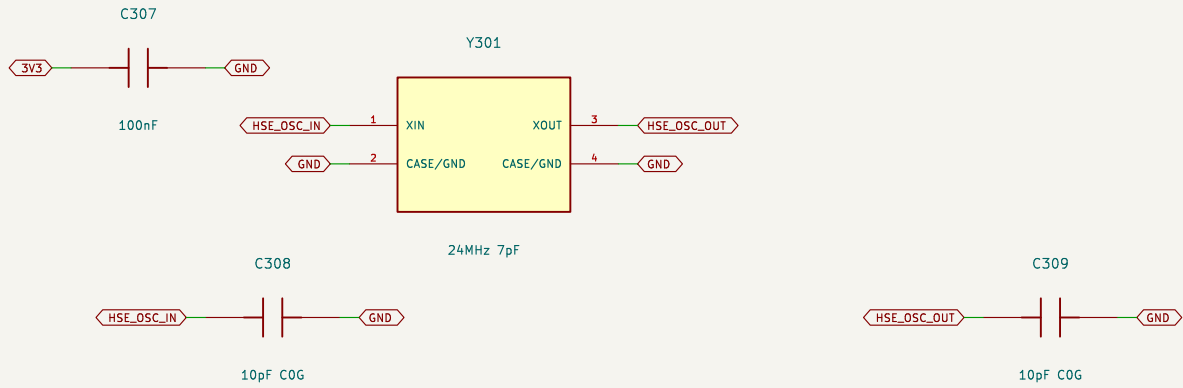
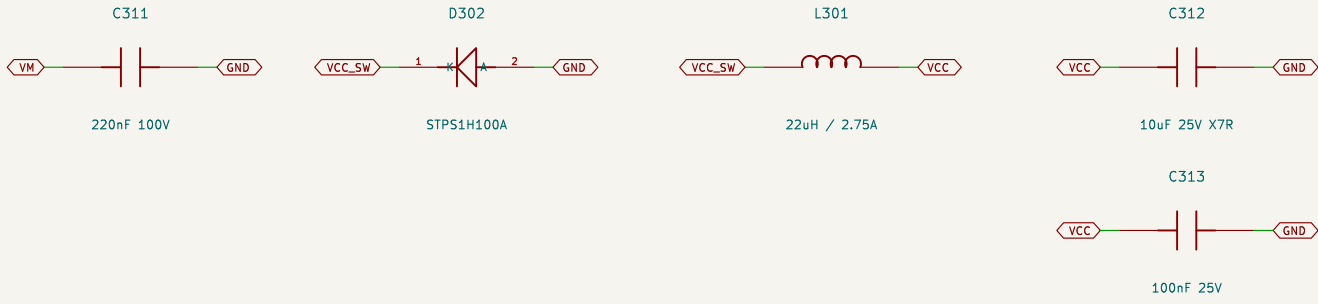
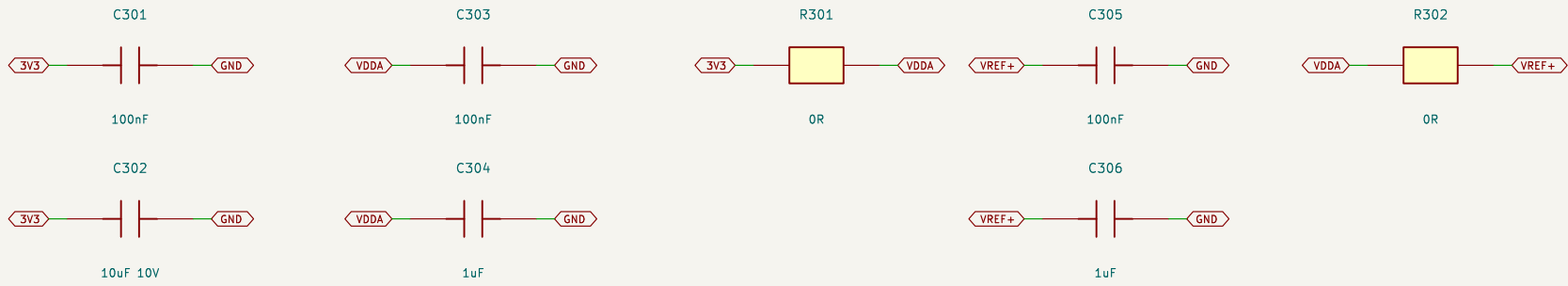
GND is common; preserve Kelvin and power–return routing
PCB layout release; prototype ratings require hardware validation
ShiroFOC

Sheet: /02 - Auxiliary Power, USB and Source Mux/
File: 02_Aux_Power_USB.kicad_sch

Title: 02 – Auxiliary Power, USB and Source Mux

Size: A2	Date: 2026-09-15	Rev: A0
KiCad E.D.A. 9.0.4		Id: 3/11

STSPIN32G4 CORE – EXTERNAL 3V3 MODE

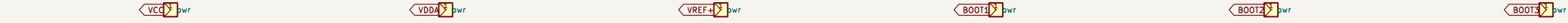


VCC BUCK: starts at 8 V. Firmware must set 10 V through the internal power-management interface after every VM return.

Pin 64 REGIN and pin 1 REG3V3/VDD are both tied to external 3V3. The internal 3V3 regulator stays disabled.

SCREF approx. 0.30 V controls VDS fault detection; it is not a precision current limit. Configure COMP1/2/4 + DAC + TIM1 break before enabling PWM.

PB8/BOOT0 has a 10k pulldown. FEEDBACK_ENABLE_N is pulled high, so the mux cannot drive PB8 while reset is active.



ERC sources: connector input / passive-filtered regulator outputs / Internal bootstrap supplies.

GND is common; preserve Kelvin and power-return routing
PCB layout release; prototype ratings require hardware validation
ShiroFOC

Sheet: /03 – STSPIN32G4 Core and Gate-Driver Supply/
File: 03_STSPIN_Core.kicad_sch

Title: 03 – STSPIN32G4 Core and Gate-Driver Supply

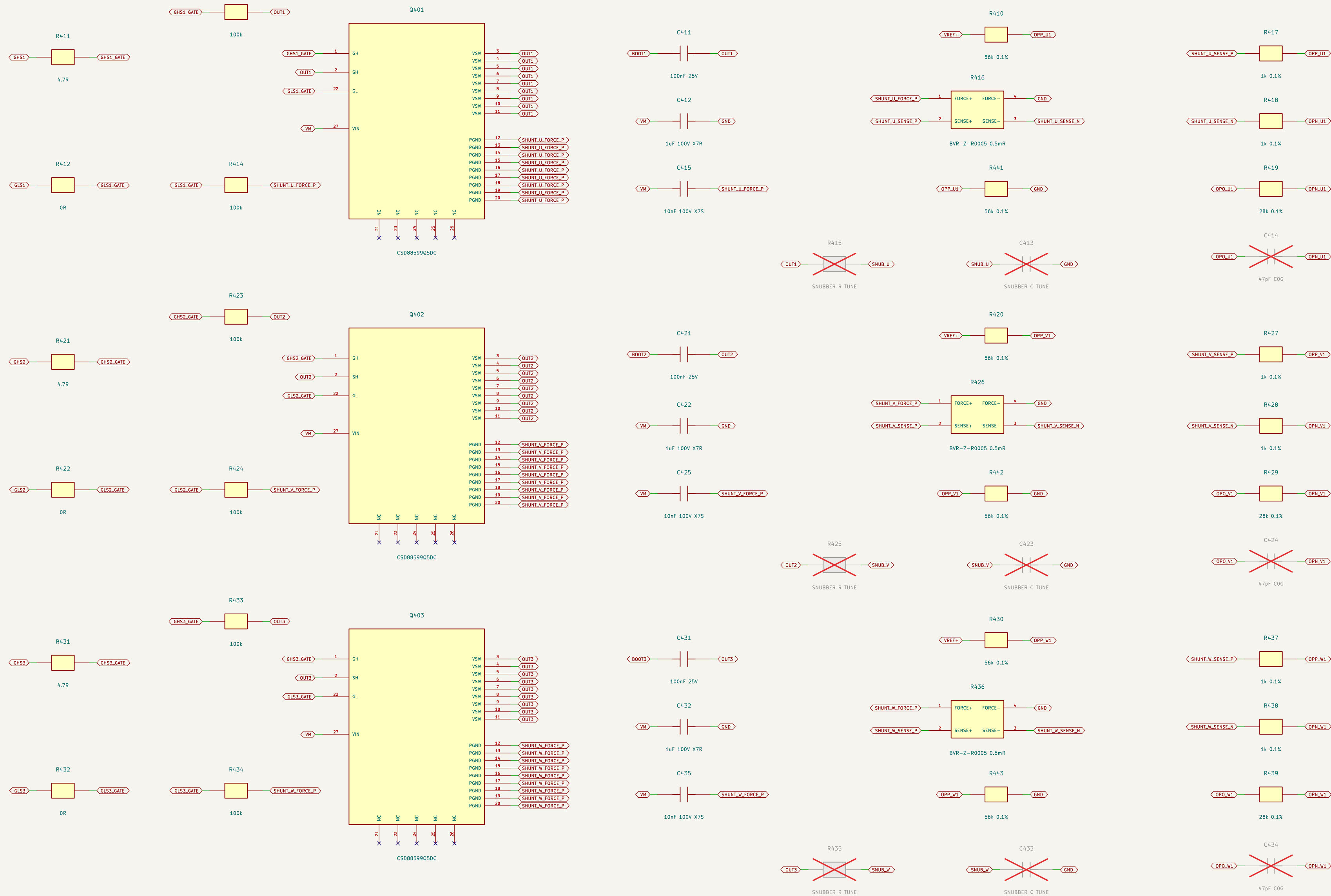
Size: A2 Date: 2026-09-15

Rev: A0

KiCad E.D.A. 9.0.4

Id: 4/11

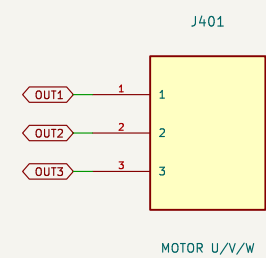
THREE-PHASE POWER STAGE – 3 x CSD88599Q5DC / 3 x 0.5 mR KELVIN SHUNTS



LAYOUT: each gate resistor sits at its MOSFET gate. Kelvin sense traces leave pads 2/3 of each BVR shunt independently and never share force copper.

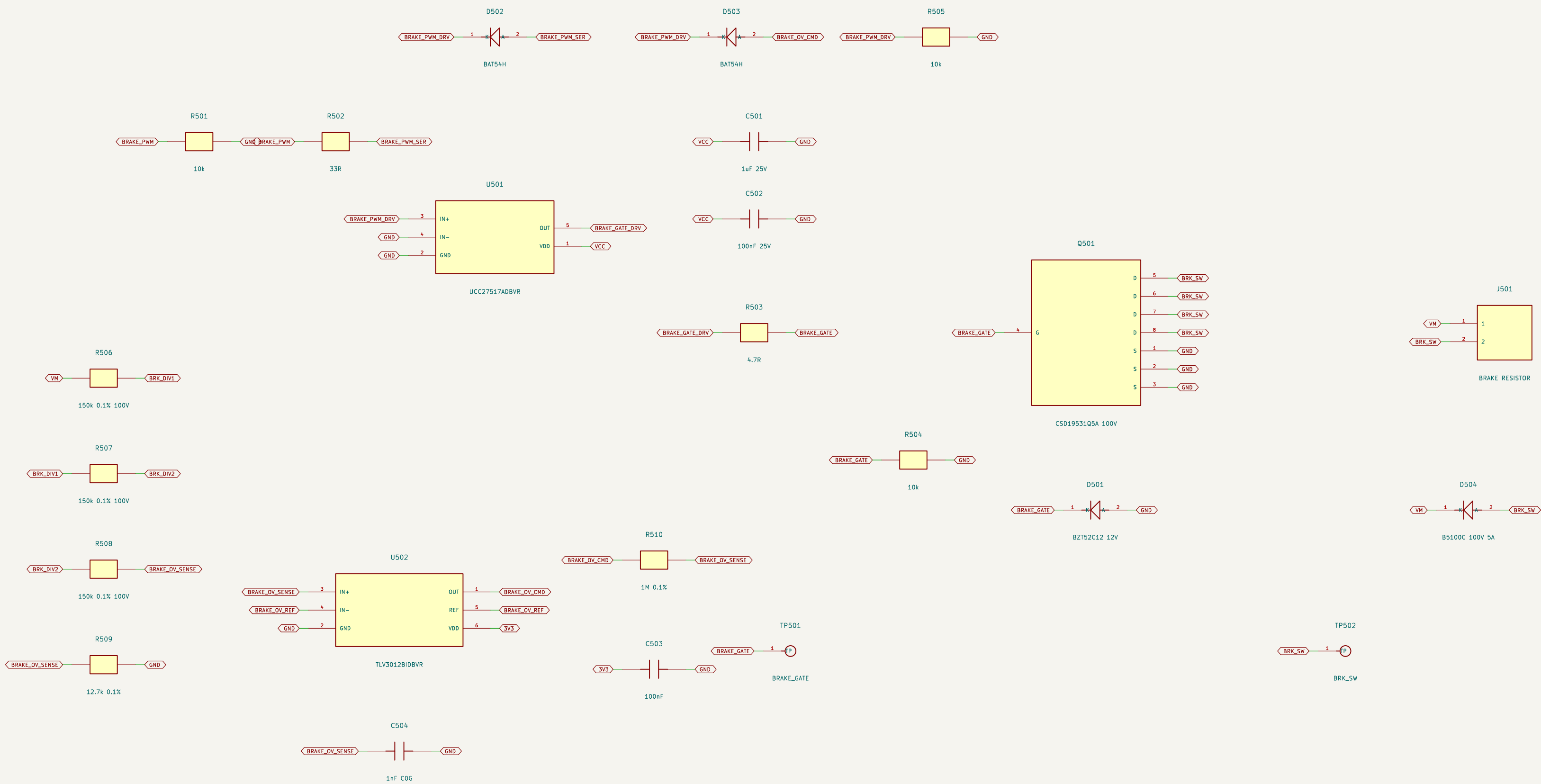
Current transfer: $V_{OUT} = V_{REF}/2 + 28 \times (SENSE_P - SENSE_N)$; 14 mV/A. Configure COMP1/2/4 on PA1/PA7/PB0 for hardware TIM1 break.

Fit GH=4.7R and GL=0R initially per TI guidance. Snubbers and 47 pF feedback capacitors are DNP tuning positions; bootstrap diodes are internal.



GND is common; preserve Kelvin and power-return routing		
PCB layout release; prototype ratings require hardware validation		
ShroFOC		
Sheet: /04 – Three-Phase Inverter and Current Sensing/ File: 04_Inverter_CurrentSense.kicad_sch		
Title: 04 – Three-Phase Inverter and Current Sensing		
Size: A2	Date: 2026-09-15	Rev: A0
KiCad E.D.A. 9.0.4		Id: 5/11

LOW-SIDE BRAKE CHOPPER – EXTERNAL RESISTOR



CSD19531Q5A: 100 V, 5.3 mR typ. / 6.4 mR max. at 10 V, 37 nC typ. Qg. The external resistor value and pulse-energy rating are application-specific.

R501/R504/R505 hold the chopper off. U502 can command it independently at approx. 45.9 V rising / 44.2 V falling; fit the external resistor before regeneration.

Connector pin 1 is raw VM. Both brake-resistor wires carry hazardous bus voltage. Place driver decoupling at U501 and use a short Kelvin source return.

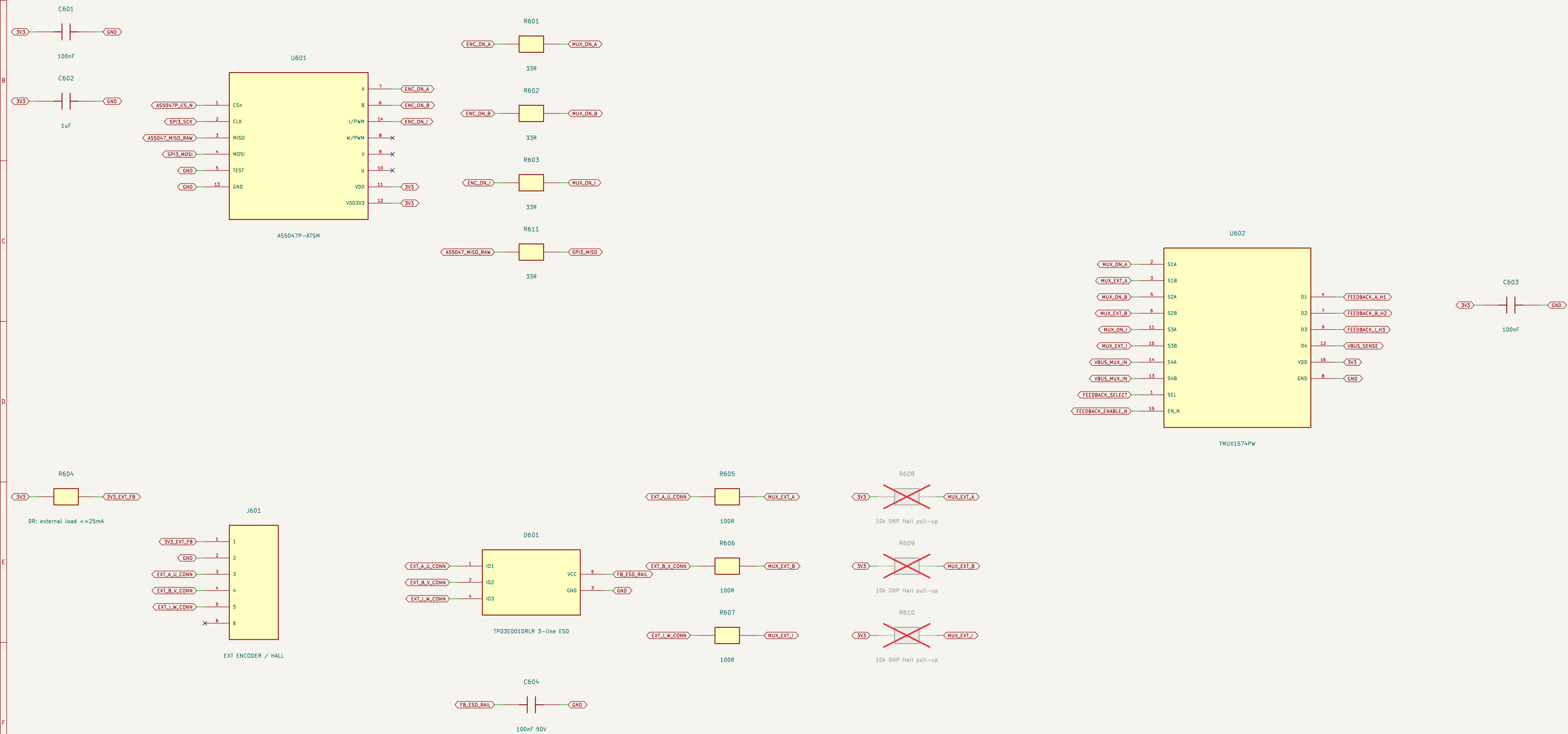
GND is common; preserve Kelvin and power–return routing
PCB layout release; prototype ratings require hardware validation
ShiroFOC

Sheet: /05 - Brake Chopper/
File: 05_Brake Chopper.kicad_sch

Title: 05 – Brake Chopper

Size: A2	Date: 2026-09-15	Rev: A0
KiCad E.D.A. 9.0.4		Id: 6/11

AS5047P + EXTERNAL ABI/HALL -> TMUX1574 -> TIM4 CH1/2/3



Default: FEEDBACK_ENABLE_N high disables all outputs; FEEDBACK_SELECT low selects onboard encoder after firmware explicitly enables the mux.

MUX channel 4 isolates VBUS_SENSE when unpowered. Enable mux and wait 2 ms before reading VM; disable PWM before disabling mux. J601 pin 6 NC.

D601 pin 5 uses an isolated 100 nF ESD rail; no DC tie to 3V3. External signals: 0–3.3 V only, including when board power is off.

GND is common; preserve Kelvin and power–return routing
PCB layout release; prototype ratings require hardware validation

ShiroFOC

Sheet: /06 – Onboard and External Position Feedback/
File: 06_Position_Feedback.kicad_sch

Title: 06 – Onboard and External Position Feedback

Size: A2

Date: 2026–09–15

Rev: A0

KiCad E.D.A. 9.0.4

Id: 7/11

BMI323 4-WIRE SPI + THREE IDENTICAL NTC CHANNELS

Pin 1 SDO=MISO, pin 14 SDI=MOSI, pin 13 SCK, pin 12 CSB. Mark package X/Y axes on silkscreen; avoid vias under package.

NTC divider behavior: open thermistor -> ADC near 3.3 V; short -> ADC near 0 V. Firmware treats both as faults and uses the hottest valid reading.

Initial fixed resistor 4.7k improves resolution over roughly 40–120 C. Calibrate thresholds against measured MOSFET case/junction temperature.

GND is common; preserve Kelvin and power–return routing		
PCB layout release; prototype ratings require hardware validation		
ShiroFOC		
Sheet: /07 – IMU and Half–Bridge Temperature/		
File: 07_IMU_Temperature.kicad_sch		
Title: 07 – IMU and Half–Bridge Temperature		
Size: A2	Date: 2026–09–15	Rev: A0
KiCad E.D.A. 9.0.4		Id: 8/11

BMI323 4-WIRE SPI + THREE IDENTICAL NTC CHANNELS

Pin 1 SDO=MISO, pin 14 SDI=MOSI, pin 13 SCK, pin 12 CSB. Mark package X/Y axes on silkscreen; avoid vias under package.

NTC divider behavior: open thermistor -> ADC near 3.3 V; short -> ADC near 0 V. Firmware treats both as faults and uses the hottest valid reading.

Initial fixed resistor 4.7k improves resolution over roughly 40–120 C. Calibrate thresholds against measured MOSFET case/junction temperature.

GND is common; preserve Kelvin and power–return routing PCB layout release; prototype ratings require hardware validation ShiroFOC Sheet: /07 – IMU and Half–Bridge Temperature/ File: 07_IMU_Temperature.kicad_sch		
Title: 07 – IMU and Half–Bridge Temperature		
Size: A2	Date: 2026–09–15	Rev: A0
KiCad E.D.A. 9.0.4		Id: 8/11

BMI323 4-WIRE SPI + THREE IDENTICAL NTC CHANNELS

The schematic shows a central BMI323 IMU (U701) connected to various components:

- Power and Grounding:** Two decoupling capacitors C701 and C702 (100nF) are connected between 3V3 and GND.
- IMU Connections:** The IMU pins are connected as follows:
 - Pin 1 SDO=MISO
 - Pin 4 IMU_INT1
 - Pin 5 VDDIO
 - Pin 6 GNDIO
 - Pin 7 GND
 - Pin 14 SDI=MOSI
 - Pin 13 SCK
 - Pin 12 CSB
 - Pin 9 INT2
 - Pin 8 VDD
 - Pins 2, 3, 10, 11 are NC (No Connection).
- SPI Interface:** A 33R resistor (R714) is placed between the IMU's INT2 pin and the SPI_MISO signal line.
- NTC Channels:** Three identical channels (A, B, C) are shown, each consisting of:
 - A 4.7k 1% fixed resistor (R701, R702, R703) connected to 3V3.
 - A 10k NTC thermistor (TH701, TH702, TH703) connected to the other end of the resistor.
 - A 10nF capacitor (C711, C712, C713) connected from the NTC node to GND.
 - An NTC phase output (TP701, TP702, TP703) connected to the NTC node.

Pin 1 SDO=MISO, pin 14 SDI=MOSI, pin 13 SCK, pin 12 CSB. Mark package X/Y axes on silkscreen; avoid vias under package.

NTC divider behavior: open thermistor -> ADC near 3.3 V; short -> ADC near 0 V. Firmware treats both as faults and uses the hottest valid reading.

Initial fixed resistor 4.7k improves resolution over roughly 40–120 C. Calibrate thresholds against measured MOSFET case/junction temperature.

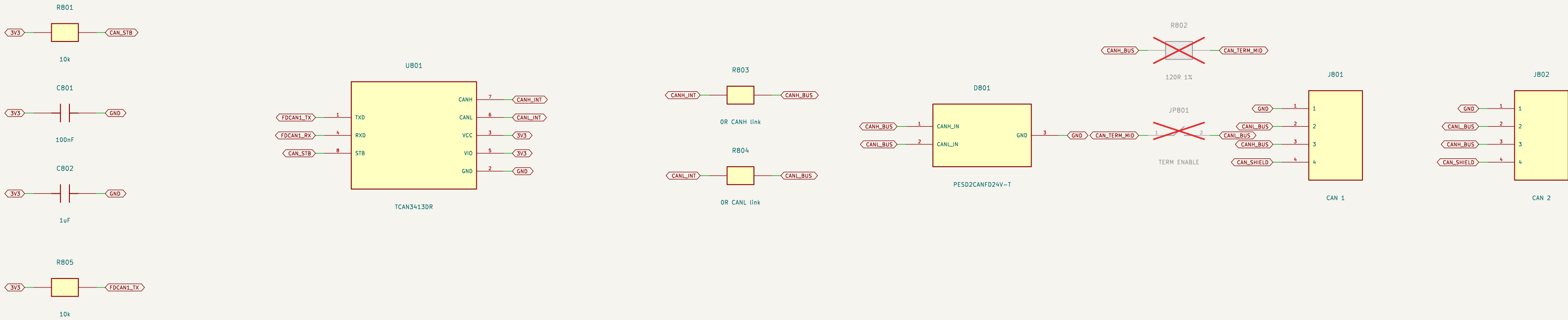
GND is common; preserve Kelvin and power-return routing PCB layout release; prototype ratings require hardware validation ShiroFOC Sheet: /07 – IMU and Half-Bridge Temperature/ File: 07_IMU_Temperature.kicad_sch		
Title: 07 – IMU and Half-Bridge Temperature		
Size: A2	Date: 2026-09-15	Rev: A0
KiCad E.D.A. 9.0.4		Id: 8/11

BMI323 4-WIRE SPI + THREE IDENTICAL NTC CHANNELS

The schematic shows a central BMI323 IMU (U701) connected to various components:

- Power and Grounding:** Two decoupling capacitors C701 and C702 (100nF) are connected between 3V3 and GND.
- IMU Connections:** The BMI323 has pins for SDO, INT1, INT2, VDDIO, GND, and SPI signals (SDI, SCK, CSB, MISO).
- NTC Channels:** Three identical channels (A, B, C) are shown. Each channel includes:
 - A 4.7k 1% resistor (R701, R702, R703) connected to 3V3 and the raw NTC signal.
 - A 10k NTC thermistor (TH701, TH702, TH703) connected to the raw signal and GND.
 - A 10nF capacitor (C711, C712, C713) connected to the phase signal and GND.
 - A pull-up resistor (TP701, TP702, TP703) connected to the phase signal and the raw NTC signal.
- Annotations:**
 - "Pin 1 SDO=MISO, pin 14 SDI=MOSI, pin 13 SCK, pin 12 CSB. Mark package X/Y axes on silkscreen; avoid vias under package."
 - "NTC divider behavior: open thermistor -> ADC near 3.3 V; short -> ADC near 0 V. Firmware treats both as faults and uses the hottest valid reading."
 - "Initial fixed resistor 4.7k improves resolution over roughly 40-120 C. Calibrate thresholds against measured MOSFET case/junction temperature."

3.3 V CAN-FD TRANSCEIVER WITH DAISY-CHAIN CONNECTORS



CAN_STB (PB10) has a pull-up: standby through reset. Configure FDCAN TX recessive before driving STB low. Fit termination only at bus ends.

CAN connector pinout is identical on both JST-GH headers. Twist CANH/CANL. Pin 4 is shield/drain provision and is not a power conductor.

GND is common; preserve Kelvin and power-return routing
PCB layout release; prototype ratings require hardware validation

ShiroFOC

Sheet: /08 - CAN / CAN-FD Interface/

File: 08_CAN.kicad.sch

Title: 08 - CAN / CAN-FD Interface

Size: A2

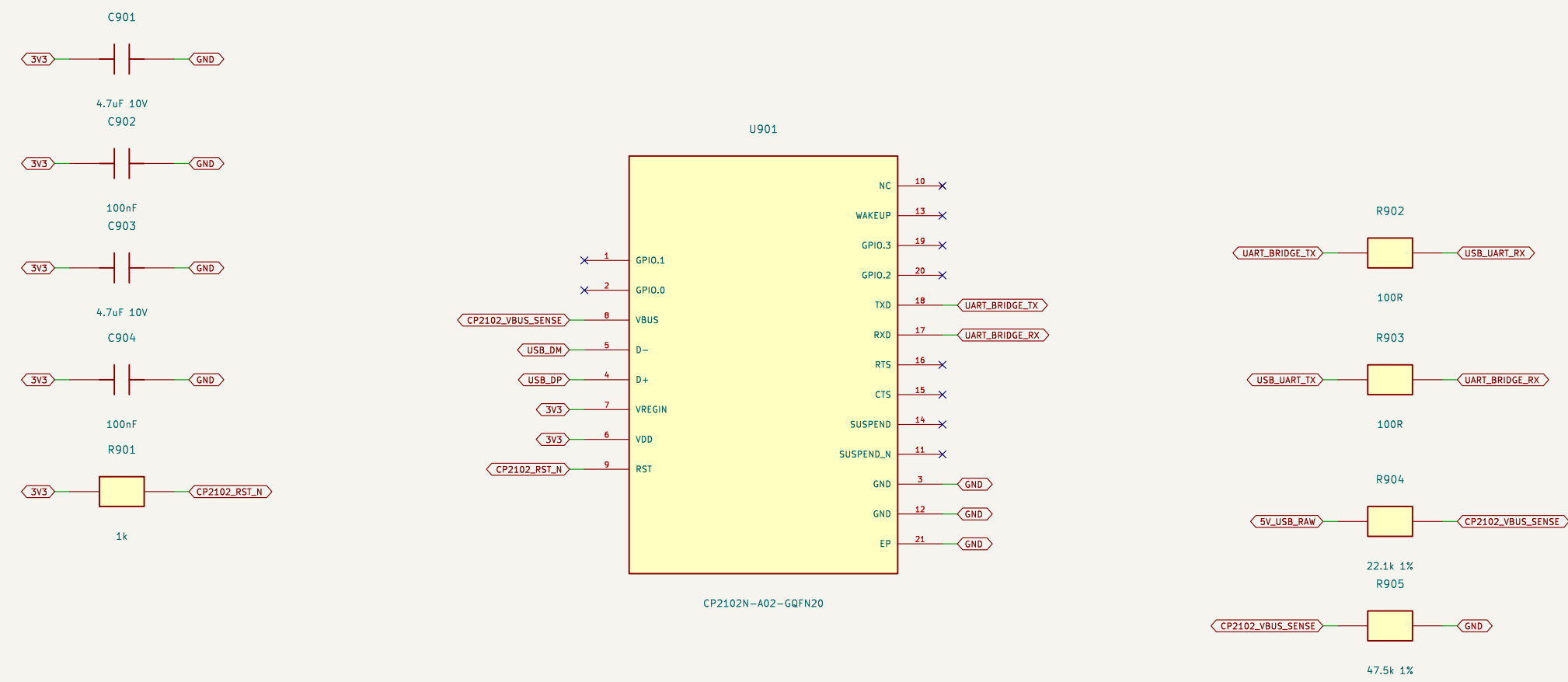
Date: 2026-09-15

Rev: A0

KiCad E.D.A. 9.0.4

Id: 9/11

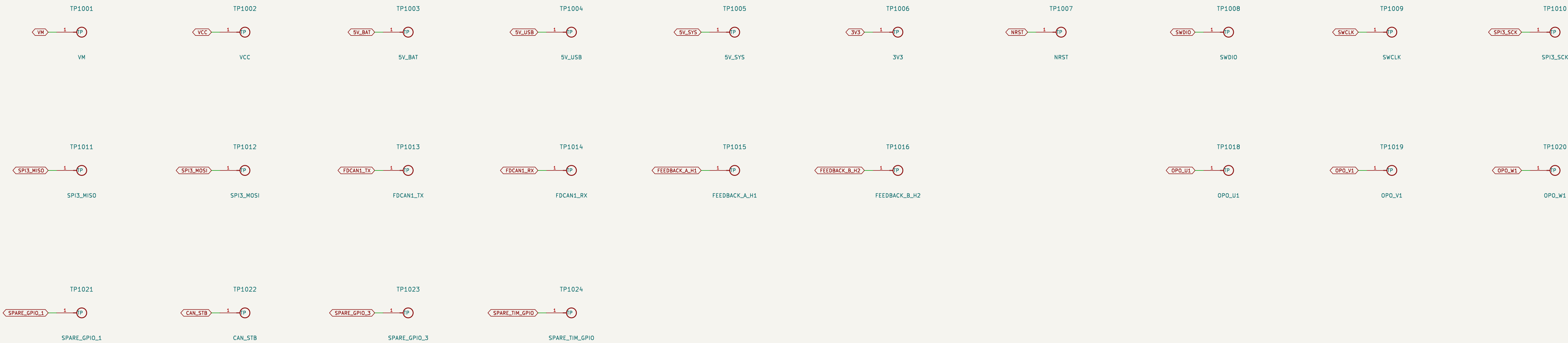
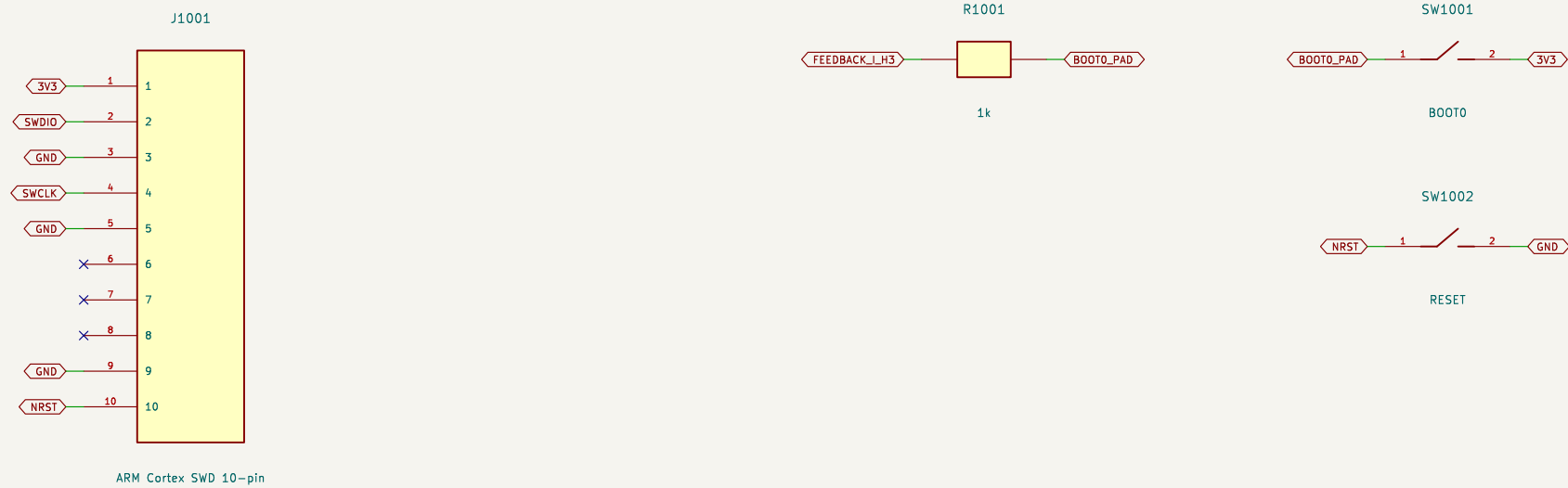
CP2102N USB-UART – SERVICE, LOGS AND ROM-UART PROGRAMMING



CP2102N regulator bypass: VREGIN and VDD both use 3V3 (datasheet Fig. 2.3). MCU and UART therefore have the same startup rail.

Manual BOOT0 and NRST remain on the SWD/test sheet. CP2102 RTS is intentionally not wired to reset in Rev A.

10-PIN ARM CORTEX SWD + BOOT/POWER TEST ACCESS



The debugger must sense target 3V3 on VTREF; it must not source 3V3 into the board. PB3 is SPI3_SCK, therefore SWO is NC in Rev A.

BOOT0 shares PB8 with feedback I/H3. The mux is disabled during reset; the 1k button path permits deliberate ROM-boot entry without hard-shortening an active encoder output.

GND is common; preserve Kelvin and power-return routing
PCB layout release; prototype ratings require hardware validation

ShiroFOC

Sheet: /10 - SWD, Boot and Test Access/

File: 10_SWD_Test.kicad_sch

Title: 10 - SWD, Boot and Test Access

Size: A2

Date: 2026-09-15

Rev: A0

KiCad E.D.A. 9.0.4

Id: 11/11